

Matthew Millward

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EDUCATION

Durham University

Predicted Graduation Date: 2026

Bachelor of Science in Mathematics (Second-Year Student)

- First Class Honours (**85.6% average**).
- Relevant modules: Programming I, Linear Algebra I, Computational Tools for Engineers and Scientists, Data Science and Statistical Computing II, Mathematical Modelling II, Statistical Inference II.

Sir Graham Balfour

2016 - 2023

*A-Levels: A*A*A*A*A in Maths, Chemistry, Physics, Further Maths and an EPQ.*

GCSEs: 9/9/9/9/9/8/8/7 (incl. Maths, English, Science, Computer Science).

PROFESSIONAL WORK EXPERIENCE

Accounting Intern, Mercedes-Benz Stourbridge

October 2022

- Conducted analysis of financial data for the Midlands Mercedes branch using proprietary software. Utilised advanced Excel functions and internal reporting tools to streamline data processing and improve accuracy.
- Enhanced data handling and reporting skills in a fast-paced, detail-oriented environment.

PERSONAL/ACADEMIC PROJECTS

Handwritten Number Recognition (Python, Scikit-Learn)

- Designed and implemented an SVM to classify **1,797 handwritten-digit images** with an 80/20 train-test split.
- Achieved over a **96% classification accuracy** by fine-tuning hyperparameters (such as gamma).
- Evaluated performance using confusion matrices and displayed the misclassified digits to enhance model precision.

Atmospheric Carbon Prediction Model (MATLAB)

- Engineered a high-performance advection-diffusion model using NASA's MERRA2 data to predict black carbon dispersion during the 2020 Australian wildfires.
- **Improved processing efficiency by 84%**, reducing data generation time from 80 seconds to 13 seconds, improving data handling techniques through MATLAB's vectorised operations and pre-allocations of arrays.

Gauge Cluster Analysis (Python, OpenCV)

- Developed a video analysis programme for real-time monitoring of 1080p gauge readings at 30fps using image pre-processing techniques (incl. colour segmentation, Gaussian blur).
- **Increased data capture rate by 90%** through machine learning-driven dynamic centre detection and leveraging adaptive colour segmentation, **reducing inconsistencies by 40%**.
- Achieved near real-time processing with an analysis time of 2.5 seconds per frame, improving operational efficiency for continuous video analysis.

Optimised Truss Structure Design (MATLAB)

- Applied optimisation algorithms (*fmincon*, *simulannealbnd*) to design lightweight trusses within stress, deflection and size constraints.
- **Reduced truss mass by 98.9%** (from 2,078 to 22kg), balancing optimisation speed and accuracy over 2000 iterations.
- Demonstrated the trade-offs between solution precision and computational time by comparing these optimisation methods and presenting my findings in a lab-report format.

DurHack Hackathon (24-hour Hackathon)

- Led a team of 3 to develop a space-themed e-commerce platform inspired by Tinder, using HTML, CSS, and JavaScript for the front end, integrating **50 product data entries** from THG. .
- Implemented a back-end system with Go for hosting, creating a user login system and shopping basket feature.
- Delivered a **fully functioning prototype within 24 hours**, demonstrating **full-stack development** capabilities and strong teamwork under pressure.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, MATLAB, C, SQL, HTML, CSS.
- **Libraries/Tools:** NumPy, Pandas, Scikit-learn, OpenCV, Flask, SQLite, Git, Matplotlib, Excel.
- **Development tools:** Git (version control), Flask (web frameworks), SQL (databases).
- **Soft Skills:** Data Analysis, Software Development, Statistical Modelling, Problem-Solving, Team Collaboration.

CERTIFICATIONS

- EdX CS50x 2024 Introduction to Computer Science **August - September 2024**
- Data Structures and Algorithms by mycodeschool **September 2024**